

# Bird Species Observation Analysis

Project Blueprint & Execution Pipeline — v2

|            |                                                                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Owner      | Tanay Nagpal — Labmentrix Internship                                                                                                                                       |
| Timeline   | 7 days from assignment date                                                                                                                                                |
| Stack      | Python / pandas · SQLite · Plotly · Streamlit · Git + GitHub · VS Code                                                                                                     |
| Deployment | GitHub repository to Streamlit Community Cloud                                                                                                                             |
| Theme      | "Dawn Chorus" — locked. Full palette in Section 4.                                                                                                                         |
| Status     | Planning complete. 13 data decisions locked, theme locked, 10 tabs specified. Ready to build.                                                                              |
| How to use | This is the controlling document. Work top to bottom. Section 3 (guardrails) is not optional — it is the difference between a correct project and a confidently wrong one. |

## 1. What We Are Delivering

| # | Deliverable                                              | Format            |
|---|----------------------------------------------------------|-------------------|
| 1 | Cleaned dataset with documented cleaning steps           | CSV + SQLite      |
| 2 | Source code for cleaning, analysis and visualisation     | Python, commented |
| 3 | Interactive dashboard, 10 tabs, deployed publicly        | Streamlit         |
| 4 | Written report of approach, findings and recommendations | PDF + in-app tab  |
| 5 | SQL query file answering the business questions          | queries.sql       |

### *Findings the pipeline must reproduce*

- **Headline:** forest records at-risk species at ~6× the grassland rate (3.65% vs 0.59%), consistent across all four shared parks.
- **Null result:** once effort and park are controlled, species richness does not differ between habitats. The apparent grassland advantage is a sampling artefact.
- **Specialists:** 18 grassland specialists versus 1 forest specialist among 49 well-sampled species.
- **Hotspots:** ANTI, CHOH, NACE, WOTR lead on species per session.
- **Environment:** richness falls as temperature rises; drizzle and serious disturbance suppress counts.
- **Daily rhythm:** grassland richness declines through the morning; forest stays flat.

## 2. Repository Structure

```
bird-species-observation-analysis/  
  data/raw/          # the two source .xlsx files, never edited  
  data/processed/    # birds_clean.csv, birds.db  
  data/reference/    # park_coordinates.csv  
  src/ingest.py       # read all sheets from both workbooks  
  src/clean.py        # the 13 locked decisions  
  src/features.py     # the 24 calculated fields (Section 7)  
  src/load_sql.py     # build SQLite database  
  src/monitoring.py   # period-agnostic monitoring layer  
  src/analysis.py     # guarded comparison functions  
  notebooks/01_exploration.ipynb · 02_analysis.ipynb  
  sql/schema.sql · queries.sql  
  app/streamlit_app.py · charts.py · theme.py · qa_bank.py  
  docs/cleaning_log.md · final_report.pdf  
  requirements.txt · README.md
```

## 3. Analytical Guardrails

Four rules. Each one, if broken, produces a result that looks statistically valid and is wrong. All four were derived from actual tests on this dataset.

| Rule                                                                                          | Evidence it is needed                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>G1.</b> Never compare raw counts across habitats. Use per-session rates.                   | Grassland has 594 sessions to forest's 814. Raw totals measure survey effort, not birdlife.                                                                                  |
| <b>G2.</b> Every habitat comparison is made within a park, using only ANTI, HAFE, MANA, MONO. | Pooled across 11 parks: grassland richer, 9.15 vs 8.61, $p=0.0005$ . Restricted to shared parks: 9.15 vs 9.10, $p=0.685$ — the effect vanishes. Two parks reverse direction. |
| <b>G3.</b> Use per-session, never per-plot.                                                   | Forest plots average 2.0 visits, grassland 2.96. Per-plot metrics inflate grassland by ~50%.                                                                                 |
| <b>G4.</b> Maps and charts encode adjusted rates, never raw totals.                           | Park ranking by raw species: MONO, MANA, ANTI, CHOH. By species per session: ANTI, CHOH, NACE, WOTR. A different answer.                                                     |

**Test before publishing any comparison:** would this number change if one park had been surveyed twice as often? If yes, the metric is wrong.

## 4. Visual Theme — Nature / Illustrated

A cream page with a dark green illustrated sidebar. The feel is a field guide rather than a business dashboard: light and open, with warmth and life coming from a hand-drawn illustration layer rather than from the data colours. This theme supersedes the earlier dark variant.

|                                                                                     | Hex     | Role                  | Notes                                                        |
|-------------------------------------------------------------------------------------|---------|-----------------------|--------------------------------------------------------------|
|    | #12805c | Forest series         | Deep forest green                                            |
|    | #b07d0a | Grassland series      | Dark gold                                                    |
|    | #c0392b | At-risk status        | Reserved. Never a series colour. Always paired with a label. |
|    | #f6f4ec | Page background       | Cream                                                        |
|    | #ffffff | Card surface          |                                                              |
|    | #e4e1d5 | Card border           |                                                              |
|    | #e8e5da | Gridline              | Recessive hairline                                           |
|    | #1a2620 | Primary ink           |                                                              |
|    | #5c6a61 | Secondary ink         |                                                              |
|   | #8a968d | Muted ink             | Axis and labels                                              |
|  | #16311f | Sidebar gradient top  | Fades to #1d4029                                             |
|  | #1b3a25 | KPI card gradient top | Fades to #245030                                             |
|  | #2b6b45 | Nav active            | White text                                                   |

**Validation.** Run against the cream surface #f6f4ec in light mode with all pairs: forest/grassland colour-vision-deficiency Delta E 8.7 (protan), normal-vision Delta E 18.4, both clear 3:1 contrast. The lighter green and gold steps (#1baf7a / #eda100) were tested and **rejected** — they fail the 3:1 contrast check on a light surface. Do not substitute colours without re-validating.

### Illustration layer

- Hand-drawn inline SVG throughout — no raster images. A few kilobytes, scales to any screen, and recolours from a single variable.
- Sidebar foot: a layered forest scene — mist gradient, far pines, near pines, a stag silhouette, and five birds in flight.
- Logo: a stylised feather and tree mark. Navigation and KPI cards carry a matching 24-pixel icon set.
- Depth is built from three silhouette layers at different opacities, so the scene reads at any sidebar height.

### One thing deliberately not copied from the reference design

The reference dashboard used four shades of green for four donut categories. Shades of a single hue are close to indistinguishable under colour-vision deficiency, so that is a genuine accessibility failure rather than a style choice. Green paired with gold keeps the nature feel while remaining readable. **Rule: never use shades of one hue for categorical data.**

### Creative features — decided

| Feature                             | Decision | Reasoning                                                                                                                                                                                                                                                           |
|-------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bird call audio in Species Explorer | BUILD    | Not decoration: 63% of all records are birds identified by song, so hearing the call closes a real loop with the ID_Method column. Curated set of roughly 20 Creative Commons recordings from Xeno-canto, stored locally with attribution — not 127 live API calls. |
| UI click and hover sounds           | SKIP     | Irritating on repeat use, unreliable under browser autoplay policy, and reads as unserious on a graded analytical deliverable.                                                                                                                                      |
| Ambient dawn chorus                 | OPTIONAL | Only with a clearly labelled toggle that is off by default.                                                                                                                                                                                                         |
| Animated KPI counters               | POLISH   | Numbers tick up on load. Cheap and common in good dashboards.                                                                                                                                                                                                       |
| Species silhouettes                 | POLISH   | Same SVG style as the sidebar, shown in species profiles.                                                                                                                                                                                                           |
| Time-of-day header tint             | POLISH   | Header gradient shifts between the 5am and 10am filters, tying the visual to the data.                                                                                                                                                                              |

Polish items are built only if Phase 7 finishes ahead of schedule. A working dashboard with one memorable feature beats a half-finished one with five.

## 5. Navigation Architecture

Ten tabs. Named for findings, not data types, so the navigation itself tells the story. Each tab opens with a one-sentence plain-language finding before any chart.

| #  | Tab                | Heading shown                                | Objective     |
|----|--------------------|----------------------------------------------|---------------|
| 1  | Overview           | "Forest is the stronghold for at-risk birds" | 1, 5          |
| 2  | Habitat Comparison | "Comparing habitats fairly"                  | 1, 5, 6       |
| 3  | Species Explorer   | "Which birds prefer which habitat"           | 2             |
| 4  | Where to Watch     | "Biodiversity hotspots across 11 parks"      | 3             |
| 5  | Seasonal & Daily   | "When birds are most active"                 | 6             |
| 6  | Environment        | "How weather shapes what we see"             | 6             |
| 7  | Data Quality       | "How this analysis was built"                | method        |
| 8  | Report             | "The full story, start to finish"            | deliverable 4 |
| 9  | Conclusion         | "What we found and what to do"               | all           |
| 10 | Ask AI             | "Ask anything about this data"               | all           |

### *Persistent elements*

- **Filter bar** (one row, above content, all tabs): habitat · park · date range · species search · at-risk-only toggle.
- **Sidebar**: project title, dataset facts, navigation, export buttons, method link, GitHub link, theme toggle.
- **Guardrail banner** on Overview and Habitat Comparison: "All comparisons shown per survey session and within park."
- **Limitations panel** (collapsible, Overview): single season; no land-use data; grassland covers 4 of 11 parks.
- **Downloads**: cleaned CSV, current filtered view, and the report PDF.

## 6. Chart Specifications

Seventeen charts. Form is chosen by the data's job, not by variety. No dual-axis charts anywhere; a legend is present whenever two series appear; every chart carries a tooltip and a one-line interpretation beneath it.

| #  | Tab          | Chart                        | Form             | Why this form                       |
|----|--------------|------------------------------|------------------|-------------------------------------|
| 1  | Overview     | Headline stat tiles          | Stat tiles       | Single values — not a chart         |
| 2  | Overview     | "6x" at-risk hero            | Hero number      | One number is the whole story       |
| 3  | Overview     | Most recorded species        | Stacked h-bar    | Magnitude + habitat split in one    |
| 4  | Habitat      | At-risk rate by park         | Grouped bar      | Magnitude across two categories     |
| 5  | Habitat      | Richness per session by park | Grouped bar      | Same form — comparison is the point |
| 6  | Habitat      | Survey effort by park        | Grouped bar      | Context for the two above           |
| 7  | Species      | Habitat preference           | Diverging bar    | Data has genuine polarity           |
| 8  | Species      | Specialist breakdown         | Bar              | Simple category magnitude           |
| 9  | Species      | Per-species profile          | Card + sparkline | Detail view, not comparison         |
| 10 | Where        | Park map                     | Bubble map       | Spatial — adjusted rates only (G4)  |
| 11 | Where        | Parks ranked                 | Horizontal bar   | Ranking with long labels            |
| 12 | Where        | Park × habitat coverage      | Heatmap          | Two-dimensional density             |
| 13 | Seasonal     | Richness by month            | Line             | Change over time                    |
| 14 | Seasonal     | Richness by hour             | Line             | The clean temporal story            |
| 15 | Environment  | Temperature vs richness      | Binned bar       | 1,408 raw points would be mush      |
| 16 | Environment  | Sky / wind / disturbance     | Small multiples  | Three parallel comparisons          |
| 17 | Data Quality | Observer comparison          | Bar              | Bias disclosure                     |

## 7. Calculated Fields

Twenty-four derived fields, built in `src/features.py` after cleaning and before the SQL load.

| Group     | Field                    | Definition                                                                                 |
|-----------|--------------------------|--------------------------------------------------------------------------------------------|
| Identity  | session_id               | Plot_Name + Date + Visit — the effort unit. Everything depends on this.                    |
|           | habitat                  | From Location_Type                                                                         |
|           | park_name                | Full park name from the 4-letter code                                                      |
|           | is_shared_park           | TRUE for ANTI / HAFE / MANA / MONO. Guardrail G2 filters on this.                          |
| Time      | month, month_name        | From Date                                                                                  |
|           | week                     | Computed but never charted (decision #12)                                                  |
|           | year, quarter            | For the auto-scaling monitoring layer                                                      |
|           | start_hour               | Hour from Start_Time                                                                       |
|           | time_band                | Early 5–6am / Mid 7–8am / Late 9–10am                                                      |
|           | session_duration_min     | End_Time – Start_Time                                                                      |
|           | day_of_season            | Days since 7 May 2018                                                                      |
|           |                          |                                                                                            |
| Cleaned   | sex_clean                | Blanks to "Undetermined" (decision #6)                                                     |
|           | is_at_risk               | PIF watchlist flag as boolean                                                              |
|           | is_stewardship           | Regional stewardship flag as boolean                                                       |
|           | is_flyover               | Flyover_Observed as boolean                                                                |
|           | distance_display         | Blank becomes "Flyover (n/a)" so flyovers do not vanish from distance charts (decision #7) |
| Metrics   | species_per_session      | Distinct species per session — the primary metric                                          |
|           | sightings_per_session    | Row count per session                                                                      |
|           | pct_at_risk_per_session  | Share of sightings that are watchlist species                                              |
| Species   | grassland_share_pct      | Percent of a species' sightings in grassland                                               |
| lookup    | specialist_class         | Grassland specialist / Forest specialist / Generalist                                      |
|           | total_sightings          | Used to filter to well-sampled species (at least 20)                                       |
| Binning   | temp_band, humidity_band | For binned environmental charts                                                            |
| Reference | lat, lon                 | Per park — 11 rows from public NPS sources, cited                                          |

## 8. Environment Constraints and Coding Standards

Every item below was found by auditing the actual development machine. Each one produces a failure that is silent locally and only appears at deployment, which is the worst kind. They are not style preferences.

### *Environment as built*

| Item         | Value                     | Note                                                                                                                                          |
|--------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Python       | 3.13.15 (python.org)      | NOT the Microsoft Store build - that one hangs on venv creation. Store aliases for python.exe / python3.exe / python3.13.exe are disabled.    |
| Launcher     | py                        | `python` is not on PATH outside the venv. Use `py -m venv .venv` to create; inside an activated venv `python` works normally.                 |
| Project path | D:\projects\bird-analysis | Deliberately short. The original path was 108 characters before any files, close to the 260-character Windows limit that breaks pip installs. |
| pandas       | 3.0.5                     | Major version 3, not 2. Older tutorials and Stack Overflow answers may use idioms that no longer apply.                                       |
| scipy        | REMOVED                   | Compiled DLLs blocked by Windows Smart App Control. Replaced by src/stats_helpers.py - see below.                                             |

### *The scipy replacement*

scipy is not installed and must not be added. Its compiled extensions are blocked by Smart App Control on this machine, and the same class of restriction can appear on deployment hosts. The two tests the project needs - Mann-Whitney U and Spearman rank correlation - are implemented in **src/stats\_helpers.py** using numpy and the standard library only, including an exact Student t distribution via the incomplete beta function. Both were validated against scipy on the real project data and match to machine precision: Mann-Whitney U=215458.5 p=0.000450024 and the four-park null result U=41549.5 p=0.685483 are identical to scipy's output, and Spearman rho and p agree to 1e-12 across every tested pair. Removing scipy leaves the project more robust, not less - one fewer dependency that can fail on deploy, and the statistical method is visible in our own code rather than hidden in a library.

### *Coding standards - mandatory*

| Rule                                                                                                                                                                                              | Why                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E1.</b> Every file write specifies <code>encoding="utf-8"</code> . This includes <code>open()</code> , <code>to_csv()</code> , <code>read_csv()</code> and every reportlab and markdown write. | Python on Windows defaults to cp1252, not UTF-8. The bird data is pure ASCII (verified - zero non-ASCII characters in either workbook), so the cleaned CSV survives, but the cleaning log and report contain dashes and arrows that will corrupt or raise <code>UnicodeEncodeError</code> . |
| <b>E2.</b> Never use PowerShell <code>&gt;</code> to create a file a program will read. Use <code>  Out-File -Encoding ascii</code> .                                                             | PowerShell redirection writes UTF-16. pip cannot read a UTF-16 <code>requirements.txt</code> , and the failure only surfaces at deployment as a confusing dependency error.                                                                                                                 |
| <b>E3.</b> All paths built with <code>pathlib</code> , never string concatenation. All filenames lowercase.                                                                                       | Windows is case-insensitive; the Linux deploy host is not. One capital letter difference works locally and fails in production with "file not found".                                                                                                                                       |
| <b>E4.</b> Two dependency files. <code>requirements.txt</code> is runtime only (6 packages). <code>requirements-dev.txt</code> adds Jupyter.                                                      | A full pip freeze produced 119 packages including 22 Jupyter ones the app never uses. Each is another chance for the deploy to fail, and the install is far slower.                                                                                                                         |
| <b>E5.</b> Pin the Python version for the deploy host in <code>runtime.txt</code> .                                                                                                               | Without it the host picks its own default, which may not match the version the code was tested against.                                                                                                                                                                                     |
| <b>E6.</b> Commit <code>.gitattributes</code> with <code>eol=lf</code> and mark <code>xlsx</code> , <code>db</code> , <code>png</code> and <code>pdf</code> as binary.                            | Prevents CRLF/LF churn in diffs and stops git corrupting binary files through line-ending conversion.                                                                                                                                                                                       |



| Rule                                                                  | Why                                                                                                                                                                                                     |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E7.</b> Test that compiled packages actually run, not just import. | scipy imported successfully at the top level and only failed when loading a compiled submodule. pyarrow and pillow are also compiled and are Streamlit dependencies - exercise a real function on each. |

**Data note:** the Wind column contains 138 values with an embedded comma, for example "Gentle breeze (8-12 mph), leaves in motion". pandas quotes these correctly on CSV export and re-reads them correctly, but any downstream consumer that splits on commas naively will corrupt those rows.

## 9. Execution Pipeline

### Phase 0 — Environment setup

| Step | Task                         | Detail / acceptance criteria                                                                                       |
|------|------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 0.1  | Create GitHub repository     | Public, named bird-species-observation-analysis. Python .gitignore.                                                |
| 0.2  | Clone and open in VS Code    | Confirm interpreter selected.                                                                                      |
| 0.3  | Virtual environment          | py -m venv .venv then .venv\Scripts\Activate.ps1. Use `py`, not `python` - see Section 8.                          |
| 0.4  | Install dependencies         | pip install -r requirements-dev.txt. NOT a raw pip freeze - see Section 8 rule E4. scipy is deliberately excluded. |
| 0.5  | Create folder structure      | Exactly as Section 2. Commit the skeleton.                                                                         |
| 0.6  | Copy raw files to data/raw/  | Lowercase .xlsx extension. Never edit these.                                                                       |
| 0.7  | Add config files             | requirements.txt, requirements-dev.txt, runtime.txt, .gitattributes, .gitignore per Section 8.                     |
| 0.8  | Verify compiled packages run | Exercise a real function in pyarrow and pillow, not just import. Rule E7.                                          |

### Phase 1 — Ingestion

| Step | Task                         | Detail / acceptance criteria                                                             |
|------|------------------------------|------------------------------------------------------------------------------------------|
| 1.1  | Read all sheets              | pd.read_excel(path, sheet_name=None) — 11 sheets per file.                               |
| 1.2  | Concatenate within each file | Expect 8,546 forest and 8,531 grassland rows. Seven empty grassland sheets are expected. |
| 1.3  | Verify habitat tag           | Location_Type must hold one value per file.                                              |
| 1.4  | Assert row counts            | Fail loudly if counts differ from the above.                                             |

### Phase 2 — Cleaning (the 13 locked decisions)

Order matters. Reconcile schema before combining; deduplicate before any counting.

| Step | Task                    | Detail / acceptance criteria                                               |
|------|-------------------------|----------------------------------------------------------------------------|
| 2.1  | Rename species code     | NPSTaxonCode and TaxonCode both become TaxonCode. [#2]                     |
| 2.2  | Drop Previously_Obs     | Constant FALSE across all grassland rows. [#3]                             |
| 2.3  | Keep Site_Name nullable | Real for forest, blank for grassland. [#1]                                 |
| 2.4  | Keep Sub_Unit_Code      | Blank means the park has no sub-units — that pattern is information. [#5]  |
| 2.5  | Cast grassland types    | Ints, floats, booleans, datetime, time. [#9]                               |
| 2.6  | Combine the two files   | pd.concat — append rows, not merge. Expect 17,077.                         |
| 2.7  | Remove duplicates       | Grassland only: 8,531 to 6,826, removing 1,705. Log before and after. [#4] |
| 2.8  | Standardise Sex         | Forest blanks to "Undetermined". Never infer sex. [#6]                     |
| 2.9  | Leave Distance nulls    | Null exactly when flyover. Structural. [#7]                                |
| 2.10 | Leave small gaps        | Scientific_Name is the species key. [#8]                                   |
| 2.11 | Write cleaning_log.md   | Every decision with row counts and reasoning. Graded deliverable.          |
| 2.12 | Export cleaned data     | birds_clean.csv — expect 15,372 rows.                                      |

### Phase 3 — Calculated fields

| Step | Task                       | Detail / acceptance criteria                                         |
|------|----------------------------|----------------------------------------------------------------------|
| 3.1  | Build all 24 fields        | Per Section 7, in src/features.py.                                   |
| 3.2  | Build park_coordinates.csv | 11 rows from public NPS sources. Cite the source in the file header. |
| 3.3  | Verify session count       | Distinct session_id must equal 1,408.                                |

## Phase 4 — SQL layer

The brief requires data stored in an SQL database for visualisation. SQLite is used because it is a single file that commits to the repository and deploys with no server or credentials, while the SQL stays standard and portable.

| Step | Task                  | Detail / acceptance criteria                                      |
|------|-----------------------|-------------------------------------------------------------------|
| 4.1  | Define schema         | sql/schema.sql — observations table plus a parks reference table. |
| 4.2  | Load cleaned data     | df.to_sql into data/processed/birds.db.                           |
| 4.3  | Index key columns     | Admin_Unit_Code, Location_Type, Scientific_Name, Date.            |
| 4.4  | Write queries.sql     | One documented query per business question (Section 10).          |
| 4.5  | Verify against pandas | Each SQL result must match the pandas equivalent.                 |

## Phase 5 — Exploratory data analysis

| Step | Task             | Detail / acceptance criteria                                                                          |
|------|------------------|-------------------------------------------------------------------------------------------------------|
| 5.1  | Distributions    | Species frequency, sightings per park, habitat, plot.                                                 |
| 5.2  | Temporal         | Monthly only — never weekly. Time-of-day is primary. [#12]                                            |
| 5.3  | Environmental    | Temperature and humidity against richness. Expect temperature rho ~ -0.2, highly significant.         |
| 5.4  | Conditions       | Expect drizzle ~7.0 and serious disturbance ~5.7 species per session against ~9.2 in good conditions. |
| 5.5  | Observer         | Report spread and balance checks. Do not correct. [#13]                                               |
| 5.6  | Detection method | Forest 5% visual vs grassland 20% — a caveat on habitat comparisons.                                  |

## Phase 6 — Guarded analysis

| Step | Task                | Detail / acceptance criteria                                                                        |
|------|---------------------|-----------------------------------------------------------------------------------------------------|
| 6.1  | Richness by habitat | Within-park, per session. Expect no significant difference. Report the null result.                 |
| 6.2  | At-risk by habitat  | Within-park, per session. Expect forest ~3.65% vs grassland ~0.59% in all four parks. The headline. |
| 6.3  | Robustness          | Re-run 6.2 separately per observer. All three must agree in direction.                              |
| 6.4  | Specialists         | Species with at least 20 sightings in shared parks. Expect 18 grassland, 1 forest, 30 generalist.   |
| 6.5  | Hotspots            | Parks and plots by species per session. Note how this differs from raw ranking.                     |
| 6.6  | Sex ratio           | Grassland only — forest has zero Female records. [#6]                                               |
| 6.7  | Monitoring layer    | Monthly now; auto-detects quarterly and yearly if more years arrive. Flag partial periods.          |

## Phase 7 — Dashboard (tabs 1–7)

| Step | Task                   | Detail / acceptance criteria                                                                                                               |
|------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1  | Theme and art modules  | app/theme.py holds the Section 4 palette; app/art.py holds the SVG illustration set. All charts and chrome read from these two files only. |
| 7.2  | App skeleton           | Tabbed layout, SQLite source, st.cache_data.                                                                                               |
| 7.3  | Filter bar and sidebar | Per Section 5.                                                                                                                             |
| 7.4  | Tabs 1–7               | Charts 1–17 per Section 6, each with its interpretation line.                                                                              |

| Step | Task                    | Detail / acceptance criteria                                                                                                                                             |
|------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.5  | Guardrail banner        | On Overview and Habitat Comparison.                                                                                                                                      |
| 7.6  | Map caveat              | Visible note: circles show survey locations and adjusted rates, not bird density.                                                                                        |
| 7.7  | Bird call audio         | Curate ~20 Creative Commons recordings from Xeno-canto for the most-recorded species. Store in app/audio/ with an attribution file. Play button on each species profile. |
| 7.8  | Polish (if time allows) | Animated KPI counters, species silhouettes, time-of-day header tint. Skip without regret if behind schedule.                                                             |

## Phase 8 — Report, Conclusion and Ask AI (tabs 8–10)

**Tab 8 — Report.** A long-form single-column scroll, not a chart grid: executive summary, dataset, cleaning, method and guardrails, findings, recommendations, limitations. The PDF is **pre-generated and committed**, not built live — the report tells one fixed story, and live generation is slow and crash-prone on the free tier. A separate "export current view as CSV" button covers the interactive case.

**Tab 9 — Conclusion.** The decision-maker's page, distinct from the Report. Each item structured as Finding then What it means then Recommended action then Confidence, grouped by business objective. The confidence column is not decoration: the at-risk finding survived every robustness check, the specialist finding is solid, the environmental correlations are real but modest, and the seasonal signal is weak and confounded. This tab also carries the honest items — the null result, the unanswerable Sustainable Agriculture objective, and the single-season limitation.

**Tab 10 — Ask AI.** Rule-based intent matching where every answer is **computed live** from the dataframe or SQL. No API key, no cost, works permanently on the free tier, and it cannot hallucinate a number. Describe it accurately as a natural-language query interface over the dataset.

| Step | Task                    | Detail / acceptance criteria                                                                                               |
|------|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 8.1  | Report tab layout       | Single column, narrative text, charts embedded where they support a point.                                                 |
| 8.2  | Generate report PDF     | reportlab. Commit to docs/final_report.pdf. Serve via st.download_button.                                                  |
| 8.3  | Conclusion tab          | Finding / meaning / action / confidence, grouped by objective.                                                             |
| 8.4  | Build qa_bank.py        | Seven intent categories per Section 11. Each intent maps to a real query.                                                  |
| 8.5  | Guardrail-aware answers | Questions like "which habitat has more birds" must return the effort-adjusted answer, never the raw count. See Section 10. |
| 8.6  | Chat UI                 | st.chat_message and st.chat_input. Show suggested questions as clickable chips.                                            |
| 8.7  | Fallback                | Unmatched questions return a helpful "here is what I can answer" list, never a guess.                                      |

## Phase 9 — Deployment

| Step | Task                      | Detail / acceptance criteria                                                      |
|------|---------------------------|-----------------------------------------------------------------------------------|
| 9.1  | Commit everything         | Including birds.db and final_report.pdf so the deployed app has them.             |
| 9.2  | Push to GitHub            | requirements.txt complete and pinned.                                             |
| 9.3  | Deploy to Streamlit Cloud | Entry point app/streamlit_app.py.                                                 |
| 9.4  | Test live                 | Every filter, tab, download and chat response. Check load time and mobile layout. |
| 9.5  | Add the live link         | To README and the final report.                                                   |

## 10. Business Questions for queries.sql

| #   | Question                                                         | Objective       |
|-----|------------------------------------------------------------------|-----------------|
| Q1  | At-risk species rate by habitat, within each shared park         | 1, 5 — headline |
| Q2  | Species richness per session by habitat, within each shared park | 6 — null result |
| Q3  | Top parks and plots by species per session                       | 3 — hotspots    |
| Q4  | Habitat preference per species, shared parks only                | 2 — specialists |
| Q5  | Which watchlist species occur where                              | 1, 5            |
| Q6  | Monthly species richness by habitat                              | 6               |
| Q7  | Richness by time-of-day band                                     | 6               |
| Q8  | Richness against temperature and humidity bands                  | 6               |
| Q9  | Richness by sky, wind and disturbance category                   | 6               |
| Q10 | Observer session counts and mean richness                        | method          |
| Q11 | Species recorded in one habitat only                             | 2               |
| Q12 | Survey effort by park and habitat                                | method          |

## 11. Ask AI — Question Bank

| Category            | Example questions                                                                                       | Answer source               |
|---------------------|---------------------------------------------------------------------------------------------------------|-----------------------------|
| Personal            | hi · hello · who are you · what can you do · thanks · bye · help                                        | Static, friendly            |
| Dataset facts       | how many species · how many records · which parks · what time period · who collected this               | Computed count              |
| Headline findings   | which habitat is better for at-risk birds · what is the main conclusion · is forest or grassland richer | Computed + guardrail text   |
| Species lookup      | tell me about [species] · is [species] at risk · where do I find [species] · most common bird           | Live query                  |
| Park lookup         | tell me about [park] · best park for birdwatching · which park has most species                         | Live query                  |
| Methodology         | why remove duplicates · why per session · why only four parks · what are the limitations                | Static, from cleaning log   |
| Guardrail-protected | which habitat has more birds · which park has the most sightings                                        | Adjusted answer — see below |

**The guardrail-protected category is the one that matters.** Asked "which habitat has more birds?", a naive bot answers "grassland, 8,531 sightings" — precisely the false conclusion the entire guardrail system exists to prevent. The required answer is: *"Raw counts favour grassland, but that reflects survey effort, not bird abundance. Adjusted per survey session and compared within parks, the habitats show no significant difference in richness — though forest holds far more at-risk species."* Built this way the chatbot defends the analysis instead of being a hole in it.

## 12. Seven-Day Schedule

| Day | Focus                        | Completion criteria                                                                                                   |
|-----|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1   | Setup, ingestion, cleaning   | birds_clean.csv exists with 15,372 rows; cleaning_log.md written.                                                     |
| 2   | Calculated fields, SQL layer | All 24 fields built; birds.db indexed; SQL verified against pandas.                                                   |
| 3   | EDA                          | Exploration notebook complete; environmental and observer patterns confirmed.                                         |
| 4   | Guarded analysis             | Headline, null result, specialists and hotspots reproduced with guardrails applied.                                   |
| 5   | Dashboard tabs 1–7           | All charts functional locally with filters working, in the Nature theme. Bird call audio wired into Species Explorer. |
| 6   | Tabs 8–10 and report         | Report tab, PDF generated and committed, Conclusion written, Ask AI answering all seven categories.                   |
| 7   | Deploy and verify            | Live on Streamlit Cloud; all links and downloads working; final review against the rubric.                            |

## 13. Risks and Mitigations

| Risk                                               | Mitigation                                                                                        |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Publishing a pooled habitat comparison by accident | Guardrail G2. Every comparison function takes a park parameter and defaults to shared parks only. |
| A chart sized by raw counts                        | Guardrail G4. All chart inputs come from rate columns.                                            |
| The chatbot contradicts the dashboard              | Rule-based only, every answer computed from the same data. No generative model.                   |
| Streamlit Cloud cannot find the database           | Commit birds.db and use relative paths throughout.                                                |
| Running out of time                                | Build Overview and Habitat Comparison first — they carry the headline. Tabs 8–10 are additive.    |
| Reviewer questions the duplicate removal           | cleaning_log.md records row counts, the 135-times-repeated example, and manager approval.         |

## Final note

The analysis is only as trustworthy as the guardrails. Two of the most natural things to do with this data — pooling all eleven parks, and comparing raw sighting counts — each produce a statistically significant result that is false. The pipeline above, the guarded analysis functions, and the rule-based chatbot are all designed so those mistakes are difficult to make by accident. Follow it in order and the headline finding will hold up to scrutiny.